

Homework 12b

Sunday, 24 March 2024 11:50 am

Homework 12B

HW 12- PART B

Exercise set 6.1 (after Thursday's lecture)

Q3, Q4, Q10, Q11, and read Examples 2 and 4 to understand the Weighted Euclidean Inner Product.

Compute $\langle \mathbf{u}, \mathbf{v} \rangle$ using the inner product in Example 7.

3.

$$(a) \quad \mathbf{u} = \begin{bmatrix} 3 & -2 \\ 4 & 8 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 & 3 \\ 1 & 1 \end{bmatrix}.$$

$$(b) \quad \mathbf{u} = \begin{bmatrix} 1 & 2 \\ -3 & 5 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 & 6 \\ 0 & 8 \end{bmatrix}$$

3. (a) $\langle \mathbf{u}, \mathbf{v} \rangle = 3(-1) - 2(3) + 4(1) + 8(1) = 3$

3. (b)

According to example 7, the inner product of $\mathbf{u}, \mathbf{v}$ is given by $Tr(\mathbf{u}^t \mathbf{v})$. Hence

$$\begin{aligned} \langle \mathbf{u}, \mathbf{v} \rangle &= Tr \left(\begin{pmatrix} 1 & 2 \\ -3 & 5 \end{pmatrix}^t \begin{pmatrix} 4 & 6 \\ 0 & 8 \end{pmatrix} \right) \\ &= Tr \left(\begin{pmatrix} 1 & -3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 4 & 6 \\ 0 & 8 \end{pmatrix} \right) \\ &= Tr \left(\begin{pmatrix} 4 & -18 \\ 8 & 52 \end{pmatrix} \right) \\ &= 56. \end{aligned}$$

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Compute $\langle \mathbf{p}, \mathbf{q} \rangle$ using the inner product in Example 8.

4.

$$(a) \quad \mathbf{p} = -2 + x + 3x^2, \quad \mathbf{q} = 4 - 7x^2$$

$$(b) \quad \mathbf{p} = -5 + 2x + x^2, \quad \mathbf{q} = 3 + 2x - 4x^2$$

(a)

According to example 8, the inner product of $a_0 + a_1x + a_2x^2, b_0 + b_1x + b_2x^2$ is given by $a_0b_0 + a_1b_1 + a_2b_2$. Hence

$$\begin{aligned}\langle -2 + x + 3x^2, 4 - 7x^2 \rangle &= (-2) \cdot (4) + (1) \cdot (0) + (3) \cdot (-7) \\ &= -8 + 0 - 21 \\ &= -29.\end{aligned}$$

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(b)

$$\begin{aligned}\langle -5 + 2x + x^2, 3 + 2x - 4x^2 \rangle &= (-5) \cdot (3) + (2) \cdot (2) + (1) \cdot (-4) \\ &= -15 + 4 - 4 \\ &= -15.\end{aligned}$$

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10. In each part, use the given inner product on $\mathbb{R}^2$ to find $\|\mathbf{w}\|$, where $\mathbf{w} = (-1, 3)$.

(a) the Euclidean inner product

(b) the weighted Euclidean inner product $\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1v_1 + 2u_2v_2$, where $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$

(c) the inner product generated by the matrix

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

(a)

Recall that the definition of Euclidean inner product of vectors $\mathbf{u} = (u_1, \dots, u_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + \dots + u_nv_n$$

and the norm generated by the product is

$$\|\mathbf{u}\| = \sqrt{u_1^2 + \dots + u_n^2}$$

Therefore

$$\|\mathbf{w}\| = \sqrt{(-1)^2 + 3^2} = \sqrt{10}$$

(b)

The norm generated by this product is

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} = \sqrt{3u_1^2 + 2u_2^2}$$

Therefore

$$\|\mathbf{w}\| = \sqrt{3(-1)^2 + 2 \cdot 3^2} = \sqrt{3 + 18} = \sqrt{21}$$

(c)

Recall that the inner product generated by matrix A can be computed as follows

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^T A^T A \mathbf{u}$$

The norm is given by

$$\|\mathbf{u}\| = \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle} = \sqrt{\mathbf{u}^T A^T A \mathbf{u}} = \sqrt{(A\mathbf{u})^T (A\mathbf{u})}$$

We have

$$A\mathbf{w} = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

Therefore

$$\|\mathbf{w}\| = \sqrt{\begin{bmatrix} 5 & 10 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \end{bmatrix}} = \sqrt{25 + 100} = \sqrt{125} = 5\sqrt{5}$$

Result

(a) $\sqrt{10}$

(b) $\sqrt{21}$

(c) $5\sqrt{5}$

11. Use the inner products in Exercise 10 to find $d(\mathbf{u}, \mathbf{v})$ for $\mathbf{u} = (-1, 2)$ and $\mathbf{v} = (2, 5)$.

We know that, for two vectors $\mathbf{u}, \mathbf{v}$, the distance is given by

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| \\ &= \|(-1, 2) - (2, 5)\| \\ &= \|(-3, -3)\|. \end{aligned}$$

(a) The Euclidean norm on $\mathbb{R}^2$ is given by

$$\|(u_1, u_2)\| = \sqrt{u_1^2 + u_2^2}.$$

Hence in this case $\|(-3, -3)\| = \sqrt{9 + 9} = 3\sqrt{2}$.

11. We have $\mathbf{u} - \mathbf{v} = (-3, -3)$.

(b) $d(\mathbf{u}, \mathbf{v}) = \|(-3, -3)\| = [3(9) + 2(9)]^{1/2} = \sqrt{45} = 3\sqrt{5}$

(c) From Problem 10(c), we have

$$[d(\mathbf{u}, \mathbf{v})]^2 = \begin{bmatrix} -3 & -3 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 13 \end{bmatrix} \begin{bmatrix} -3 \\ -3 \end{bmatrix} = 117$$

Thus

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{117} = 3\sqrt{13}$$

EXAMPLE 2 Weighted Euclidean Inner Product

Let $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ be vectors in $\mathbb{R}^2$. Verify that the weighted Euclidean inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1 v_1 + 2u_2 v_2$$

satisfies the four inner product axioms.

Solution

Note first that if $\mathbf{u}$ and $\mathbf{v}$ are interchanged in this equation, the right side remains the same. Therefore,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$$

If $\mathbf{z} = (z_1, z_2)$, then

$$\begin{aligned} \langle \mathbf{u} + \mathbf{v}, \mathbf{z} \rangle &= 3(u_1 + v_1)z_1 + 2(u_2 + v_2)z_2 \\ &= (3u_1 z_1 + 2u_2 z_2) + (3v_1 z_1 + 2v_2 z_2) \\ &= \langle \mathbf{u}, \mathbf{z} \rangle + \langle \mathbf{v}, \mathbf{z} \rangle \end{aligned}$$

which establishes the second axiom.

Next,

$$\langle k\mathbf{u}, \mathbf{v} \rangle = 3(ku_1)v_1 + 2(ku_2)v_2 = k(3u_1 v_1 + 2u_2 v_2) = k\langle \mathbf{u}, \mathbf{v} \rangle$$

which establishes the third axiom.

Finally,

$$\langle \mathbf{v}, \mathbf{v} \rangle = 3v_1 v_1 + 2v_2 v_2 = 3v_1^2 + 2v_2^2$$

Obviously, $\langle \mathbf{v}, \mathbf{v} \rangle = 3v_1^2 + 2v_2^2 \geq 0$. Further, $\langle \mathbf{v}, \mathbf{v} \rangle = 3v_1^2 + 2v_2^2 = 0$ if and only if $v_1 = v_2 = 0$ —that is, if and only if $\mathbf{v} = (v_1, v_2) = \mathbf{0}$. Thus the fourth axiom is satisfied.

EXAMPLE 4 Using a Weighted Euclidean Inner Product

It is important to keep in mind that norm and distance depend on the inner product being used. If the inner product is changed, then the norms and distances between vectors also change. For example, for the vectors $\mathbf{u} = (1, 0)$ and $\mathbf{v} = (0, 1)$ in $\mathbb{R}^2$ with the Euclidean inner product, we have

$$\|\mathbf{u}\| = \sqrt{1^2 + 0^2} = 1$$

and

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \|(1, -1)\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

However, if we change to the weighted Euclidean inner product of Example 2,

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1 v_1 + 2u_2 v_2$$

then we obtain

$$\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = [3(1)(1) + 2(0)(0)]^{1/2} = \sqrt{3}$$

and

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}) &= \|\mathbf{u} - \mathbf{v}\| = \left\{ (1, -1), (1, -1) \right\}^{1/2} \\ &= [3(1)(1) + 2(-1)(-1)]^{1/2} = \sqrt{5} \end{aligned}$$